Q11: Why Do LLMs Hallucinate?

Paradise (nuc13)

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The question

- "Why do models hallucinate, and can they know when they are unsure?"
- A model output that is fluent, confident, and false — where does it come from?
- Answer preview: **three levels of one mechanism — truth is a narrow subclass of plausible, calibrated predictors must spread mass over the unattested, and rare facts lose the argmax to popular continuations**

Hallucination $\neq$ lying

- Lying = intent to deceive: a model of the reader + a goal
- A next-token predictor has neither; it *samples from a distribution*
- Fluent falsehood is an epistemic failure, not a moral one
- Taxonomy for tonight:
  - **Fabrication**: no source exists (invented paper, invented API)
  - **Misattribution**: real entities, false combination (wrong venue, wrong year)
  - **Faithful error**: right premises, slipped reasoning (arithmetic slip)

Account 1: compression / generalization

- Pre-training = lossy compression under a parameter budget
- The objective rewards *plausible continuation*; truth is one subclass of plausible
- The corpus contains fiction, error, contradiction $\Rightarrow$ a perfect predictor of it still emits falsehoods ("imitative falsehoods" — TruthfulQA; best early model 58% truthful vs. 94% human, *larger was worse*)
- QA accuracy tracks pre-training frequency; long-tail facts are not rescued by scale

Account 2: calibration

- Generation *samples* from a distribution — hallucination = sampling beyond the reliability boundary
- Kalai–Vempala (STOC 2024): for *arbitrary* facts (truth not inferable from data), any calibrated model with bounded per-fact probability must hallucinate at a rate $\approx$ its **monofact rate** (fraction of facts seen exactly once — a Good–Turing estimate)
- Hedges: arbitrary facts only; ideal data; pre-training only — post-training trades calibration for caution

Account 3: knowledge boundaries

- Two stores: **parametric memory** (weights) vs. **context** (prompt, retrieval)
- Hallucination lives at their conflict: context implies A, parameters insist on B
- Facts localize in mid-layer MLPs; rank-one edits rewrite them (ROME) — recall is a lookup that can fail independently of fluency
- Fine-tuning on *new* knowledge: learned slowly, and raises hallucination linearly as it is memorized (Gekhman et al. 2024) $\Rightarrow$ SFT teaches the *form* of answers, not new facts

Mitigations, honestly

- **Retrieval (RAG)**: extends what counts as attested; reduces hallucination in dialogue — but the retriever can fetch wrong passages, generators can ignore right ones, and retrieval *hurts* on popular entities the model already knows (Mallen et al.
  2023)
- **Post-training**: reduces hallucination, degrades calibration (overconfident refusal/hedge rates)
- **Decoding / verification**: filters samples; cannot move the boundary for unattested facts
- No cure — each mitigation swaps failure modes; pick the signature you can audit

Evaluation

Instrument	**What it actually measures**
TruthfulQA	resistance to *imitative* falsehoods (popular wrong answers)
Factual-association probes	parametric *storage* (cloze + causal edit), not end-to-end truthfulness
HaluEval	*recognition* of hallucination in context (35K pairs; models far from ceiling)

No single number is "the hallucination rate" — track the slice matching your failure mode.

Toy demo (NumPy, 100% truthful corpus)

- Smoothed count model, Zipf-popular facts: greedy accuracy by training count
- 0.00 (count 0–2) $\rightarrow$ 0.44 (3–5) $\rightarrow$ 1.00 (6+): **rare facts lose the argmax to the most popular entity**
- Median P(truth) crosses 0.5 only at count 16+: sampling hallucinates on a *strictly larger* set than greedy
- Monofact rate 4.0% $\approx$ empirical missing-fact rate 3.1% (Good–Turing flavor)
- *Fluency is free; truth is not* — route around it: retrieval for the unattested, abstention for the unsure, audit for the rest

Reading

- Ji et al.
  2023 — the NLG hallucination survey (definitions, taxonomy)
- Kalai & Vempala 2024 — calibrated models must hallucinate (the theorem)
- Lin et al.
  2022; Gekhman et al.
  2024 — imitative falsehoods; fine-tuning on new knowledge
- Lewis et al.
  2020; Shuster et al.
  2021; Mallen et al.
  2023 — RAG, its gains, its failure modes